Envelope Tilings: notes and references

Companion notes for the editor at zevahcle.github.io/envelope-tilings · source and standalone file at github.com/zevahcle/envelope-tilings · 3 November 2026

1. What the editor shows

Envelope mode. Take a polygon and glue two copies of it along the boundary: a "pillowcase" or *dihedron*. Cut the front sheet along any curves, keep each flap attached to the back sheet along one straight segment of the boundary (its hinge), and open the flaps flat. Opening a flap reflects it across its hinge, so the result is a planar region (the *net*) of twice the original area. For a **rectangle**, and for the three triangles with angles **45°–45°–90°**, **30°–60°–90°** and **60°–60°–60°**, *every* such net tiles the plane. This is Akiyama's *tile-maker theorem* [1, 4, 5]; for the rectangle, Akiyama and Matsunaga proved the stronger statement that every net satisfies the Conway criterion and therefore tiles by translations and half-turns [2, 3]. Langerman and Winslow [6] gave the reason in one line: a closed surface is a tile-maker exactly when every one of its cone points has angle $360°/m$ for an integer m, and these four dihedra are the only convex ones.

The editor shows this directly, and it also shows the two ways the construction fails for other polygons. First, a flap can fail to *flatten*: for a non-convex polygon such as the hat, the reflected flap may land on the back sheet (drawn in red). Second, a net may flatten and still not tile. For non-tile-makers the editor searches for a Conway decomposition of the net and tiles with it when one exists (the parallelogram with a side-to-side cut, or the regular hexagon with no cut), and otherwise reports that no such tiling was found.

Seam mode. The same idea, read differently: an envelope cut is a curve drawn on a pair of matched edges of a tiling, and it keeps the tiling as long as the two edges it replaces still fit together. For a general tiling by one tile, every edge of the tile meets some edge of a neighbour either directly, by a half-turn about the contact midpoint, or through a reflection. Following these contacts groups the edges of the tile into classes; one drawn curve determines the curve on every edge of its class, and a cycle of contacts can force the curve to be centrally symmetric, mirror symmetric, or straight. For lattice tiles this is the classical Escher-style edge modification [13, 14, 15]. For the **Spectre** all fourteen edges form a single free class whose curves alternate in orientation, exactly the construction in the chiral monotile paper [8, Lemma 2.1]. For the **hat** and the whole family Tile(a,b) [7] every class contains a mirrored contact, so every edge is forced straight: the hat admits no edge-curving at all, and the only freedom is the two side lengths a, b.

2. What is known and what is new

The envelope observation, its classification, the Conway-tile description of the nets, and the generalisation to flat closed surfaces are all published by Akiyama and his coauthors [1–6]. Replacing the edges of the Spectre by curves is part of the original construction [8], and the hat's authors describe such changes as "trivial modifications of tiling edges"; the dynamical-systems literature treats them as a change that preserves mutual local derivability [9, 10, 11].

What we have not found in print is the edge-by-edge contact analysis for the hat and its family (the classes of its fourteen edges and the rigidity that follows), the hat as an explicit example of why the tile-maker hypothesis is needed, the reformulation of the envelope cut as a seam deformation, and an interactive editor doing all of this. The rigidity statement is checked by computer on substitution patches of the hat, turtle and Spectre; the code and tests are in the repository.

3. References

Bibliographic data were checked on 3 November 2026 against Crossref and the publishers' pages; every DOI below resolves.

[1] J. Akiyama, *Tile-makers and semi-tile-makers*, American Mathematical Monthly 114 (7) (2007) 602–609. doi:10.1080/00029890.2007.11920450

Defines tile-makers (polyhedra and dihedra all of whose nets tile) and classifies the convex ones: isosceles tetrahedra and the doubly covered rectangle, equilateral, isosceles right and 30°–60°–90° triangles.

[2] J. Akiyama, K. Matsunaga, *Unfoldings of an envelope*, European Journal of Combinatorics 80 (2019) 3–16. doi:10.1016/j.ejc.2018.02.023

Cutting a doubly covered rectangle along any spanning tree of its corners gives a Conway tile; two unfoldings along non-crossing trees are hinged dissections of each other; conditions for a Conway tile to fold back into an envelope.

[3] J. Akiyama, K. Matsunaga, *An algorithm for folding a Conway tile into an isotetrahedron or a rectangle dihedron*, Journal of Information Processing 28 (2020) 750–758. doi:10.2197/ipsjjip.28.750

The converse direction: every Conway tile folds into an isotetrahedron or a rectangle dihedron, with an algorithm.

[4] J. Akiyama, K. Matsunaga, *Treks into Intuitive Geometry: The World of Polygons and Polyhedra*, 2nd ed., Springer Nature Singapore, 2024. doi:10.1007/978-981-99-8608-8 (1st ed., Springer Japan, 2015, doi:10.1007/978-4-431-55843-9)

The book in which the pillowcase appears. Chapter 2, "The tile-maker theorem and its applications to art and designs" (2nd ed. pp. 35–88), proves the dihedral tile-maker theorem and extends it to flat closed surfaces; Chapter 5, "Reversible pairs of figures" (pp. 177–264), contains "Envelope magic" and its trapezoidal and generalised versions; Chapter 6, "Reversibility and foldability of Conway tiles" (pp. 265–344), folds strips into rectangular dihedra.

[5] J. Akiyama, K. Hirata, *On convex developments of a doubly-covered square*, in: Combinatorial Geometry and Graph Theory (IJCCGGT 2003), Lecture Notes in Computer Science 3330, Springer, 2005, pp. 1–13. doi:10.1007/978-3-540-30540-8_1

Early study of the nets of the doubly covered square.

[6] S. Langerman, A. Winslow, *A complete classification of tile-makers*, Japan Conference on Discrete and Computational Geometry and Graphs (JCDCGG), 2015. andrewwinslow.com/papers/tilemakers-jcdcgg15.pdf

A closed surface is a tile-maker if and only if every point has angle $360°/m$; the nets of tile-makers are exactly the prototiles of isohedral tilings using at least two rotations and no reflections. Hence no closed surface can be an "aperiodic tile-maker".

[7] D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss, *An aperiodic monotile*, Combinatorial Theory 4 (1) (2024), article 6. doi:10.5070/C64163843 · arXiv:2303.10798

The hat, the family Tile(a,b) (the 13-gon treated as a 14-gon, Section 6), the H, T, P, F metatiles and the substitution used by the editor. Every hat tiling uses reflected hats; the metatile decomposition bounds their proportion by one quarter.

[8] D. Smith, J. S. Myers, C. S. Kaplan, C. Goodman-Strauss, *A chiral aperiodic monotile*, Combinatorial Theory 4 (2) (2024). doi:10.5070/C64264241 · arXiv:2305.17743

The Spectre: Tile(1,1) with every edge replaced by alternating copies of one curve, or by a curve with half-turn symmetry (Lemma 2.1), which forbids reflections. The authors call different curve choices "trivial modifications of tiling edges".

[9] M. Baake, F. Gähler, L. Sadun, *Dynamics and topology of the Hat family of tilings*, Israel Journal of Mathematics 270 (2025) 449–485. doi:10.1007/s11856-025-2780-8 · arXiv:2305.05639
Shape deformations of the hat family: a four-dimensional family of linear deformations, all topologically conjugate; curving edges is a trivial change in this framework.

[10] M. Baake, F. Gähler, J. Mazáč, L. Sadun, *On the long-range order of the Spectre tilings*, Discrete & Computational Geometry 76 (2025) 739–762. doi:10.1007/s00454-025-00756-z
Spectral and dynamical properties of the Spectre tilings; the results are independent of the edge curve.

[11] A. Clark, L. Sadun, *When shape matters: deformations of tiling spaces*, Ergodic Theory and Dynamical Systems 26 (1) (2006) 69–86. doi:10.1017/S0143385705000623
The general theory of shape deformations of tilings and when they change the tiling space.

[12] J. Akiyama, E. D. Demaine, S. Langerman, *Polyhedral characterization of reversible hinged dissections*, Graphs and Combinatorics 36 (2020) 221–229. doi:10.1007/s00373-019-02041-2 · arXiv:1803.01172
“Envelope magic” as the dihedral case of “parcel magic”: reversible hinged dissections come from nets of a common polyhedron.

[13] D. Schattschneider, *Will it tile? Try the Conway criterion!*, Mathematics Magazine 53 (4) (1980) 224–233. doi:10.1080/0025570X.1980.11976859
The Conway criterion used by the editor to test nets of non-tile-makers: a boundary split into six arcs, two of them translates and the other four centrally symmetric.

[14] H. Heesch, O. Kienzle, *Flächenschluß: System der Formen lückenlos aneinanderschließender Flachteile*, Springer, Berlin, 1963. doi:10.1007/978-3-642-94883-1
The 28 Heesch types: which edge symmetries (translation, half-turn, glide) are allowed on each edge of an isohedral tile, the origin of the edge-class idea.

[15] B. Grünbaum, G. C. Shephard, *Tilings and Patterns*, W. H. Freeman, New York, 1987 (reprinted by Dover, 2016).
Chapter 6: the 81 isohedral types, with the Heesch-type edge symmetries, and the standard reference for modifying tile edges.

[16] C. S. Kaplan, *hatviz* (software, BSD-3 licence), github.com/isohedral/hatviz; the hat and Spectre pages, cs.uwaterloo.ca/~csk/hat and cs.uwaterloo.ca/~csk/spectre.
The substitution rules for hat patches in the editor are ported from hatviz; the Spectre substitution follows the chiral paper.

These notes accompany the Envelope Tilings editor (MIT licence). Comments and corrections: open an issue at github.com/zevahcle/envelope-tilings.